

A divisor of the gcd-product: a proof of Bala's conjecture on OEIS A129364

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Abstract

Let $d(n) = \prod_{k=1}^n \prod_{e|k} e^{k/e}$, OEIS A129364, and let $G(n) = \prod_{j,k \leq n} \gcd(j, k)$, OEIS A092287. In April 2007 P. Bala conjectured that $d(n)$ divides $G(n)$ for every n . We prove it, in the sharp form

$$v_p(G(n)) - v_p(d(n)) = \sum_{i \geq 1} B(\lfloor n/p^i \rfloor), \quad B(M) = \sum_{k=1}^M (M \bmod k) \quad (\text{A004125}),$$

for every prime p . Every term on the right is a sum of remainders, hence nonnegative, and the divisibility follows at once. Three ingredients: the layer-cake count for $v_p(G(n))$, Legendre's formula together with the nested-floor identity for $v_p(d(n))$, and the observation that $M^2 - \sum_{k \leq M} k \lfloor M/k \rfloor$ is exactly $\sum_{k \leq M} (M \bmod k)$.

1 The sequences and the conjecture

OEIS A129364 (Peter Bala, 11 April 2007) has offset 1 and is

$$d(n) = \prod_{k=1}^n \text{A066841}(k) = \prod_{k=1}^n \prod_{e|k} e^{k/e} = 1, 2, 6, 96, 480, 207360, 1451520, 2972712960, \dots$$

The entry also records the equivalent closed form

$$d(n) = \prod_{k=1}^n (\lfloor n/k \rfloor!)^k, \tag{1}$$

which we use throughout; the two agree because writing $k = e \cdot m$ turns $\prod_{k \leq n} \prod_{e|k} e^{k/e}$ into $\prod_{e \leq n} e^{\sum_{m \leq n/e} m}$, and the same rearrangement applied to (1) gives the same exponent of each e .

Write $G(n) = \text{A092287}(n) = \prod_{j=1}^n \prod_{k=1}^n \gcd(j, k)$. The entry carries

*Conjecture: $a(n)$ divides **A092287**(n) for all n — see comments in **A129365**.
— Peter Bala, Apr 2007*

As of the “Last modified” line on the live entry (24 August 2026) this is still recorded as an unproven conjecture, with no proof in the entry's links or programs.

Throughout p is a prime, v_p is the p -adic valuation, and $B(M) = \sum_{k=1}^M (M \bmod k)$ is A004125, with $B(0) = 0$.

2 Three ingredients

Lemma 1 (layer cake). $v_p(G(n)) = \sum_{i \geq 1} \lfloor n/p^i \rfloor^2$.

Proof. $v_p(\gcd(j, k)) = \min(v_p(j), v_p(k))$ is the number of $i \geq 1$ with $p^i \mid j$ and $p^i \mid k$. Summing over $j, k \leq n$ and exchanging the order of summation, the two divisibility conditions are independent, so level i contributes $(\#\{j \leq n : p^i \mid j\})^2 = \lfloor n/p^i \rfloor^2$. (This is the formula of A092287, confirmed on that entry by C. R. Greathouse IV in April 2013.) $\square$

Lemma 2 (nested floors). $\lfloor n/(kq) \rfloor = \lfloor \lfloor n/q \rfloor / k \rfloor$ for positive integers n, k, q .

Proof. For real x and a positive integer k , $\lfloor x/k \rfloor = \lfloor \lfloor x \rfloor / k \rfloor$; apply with $x = n/q$. $\square$

Lemma 3 (remainder identity). $M^2 - \sum_{k=1}^M k \lfloor M/k \rfloor = B(M)$ for every integer $M \geq 0$.

Proof. $M^2 = \sum_{k=1}^M M$, and $M - k \lfloor M/k \rfloor = M \bmod k$ termwise. $\square$

3 The theorem

Theorem 4. For every $n \geq 1$ and every prime p ,

$$v_p(G(n)) - v_p(d(n)) = \sum_{i \geq 1} B(\lfloor n/p^i \rfloor).$$

Proof. Put $M_i = \lfloor n/p^i \rfloor$. By (1), Legendre's formula and Lemma 2,

$$v_p(d(n)) = \sum_{k=1}^n k v_p(\lfloor n/k \rfloor!) = \sum_{k=1}^n k \sum_{i \geq 1} \lfloor \lfloor n/k \rfloor / p^i \rfloor = \sum_{i \geq 1} \sum_{k=1}^n k \lfloor n/(kp^i) \rfloor = \sum_{i \geq 1} \sum_{k=1}^{M_i} k \lfloor M_i/k \rfloor,$$

where the last step uses Lemma 2 again in the form $\lfloor n/(kp^i) \rfloor = \lfloor M_i/k \rfloor$, the terms with $k > M_i$ vanishing. Subtracting this from Lemma 1 and applying Lemma 3 at each level i ,

$$v_p(G(n)) - v_p(d(n)) = \sum_{i \geq 1} \left(M_i^2 - \sum_{k=1}^{M_i} k \lfloor M_i/k \rfloor \right) = \sum_{i \geq 1} B(M_i). \quad \square$$

Corollary 5 (Bala's conjecture). $d(n)$ divides $G(n)$ for every $n \geq 1$.

Proof. Each $M \bmod k$ is a nonnegative integer, so $B(M) \geq 0$ and hence $v_p(G(n)) \geq v_p(d(n))$ for every prime p by Theorem 4. A positive rational with nonnegative valuation at every prime is an integer. $\square$

Remark (the same statement lives on two entries). The quotient $G(n)/d(n)$ is A129365, whose own comment list opens with “*Conjectures: A) $a(n)$ is always an integer*” (Bala, 13 April 2007). That is the present conjecture restated, and the two entries have never been cross-annotated as being the same claim — A129364 merely points at “comments in A129365”. A companion note proves Conjecture A of A129365 from the same closed form; the present note is the divisibility form, stated and proved directly on A129364 so that entry stands on its own. Whoever records these should record both, and note that one proof settles both.

Remark (honest assessment). Elementary throughout: the layer-cake count, Legendre, and one floor identity. The argument is the same one that appears in the companion note on A129365, because the two conjectures are the same statement; nothing new is proved here beyond making that explicit and giving A129364 a self-contained treatment. The natural destination is a comment on the entry.

4 Verification

A verification script, written separately from this note, checks every claim and prints every number quoted. It (i) reproduces the first 12 published terms of A129364 from the defining product over divisors; (ii) confirms that the entry's two formulas agree, $\prod_{k \leq n} A066841(k) = \prod_{k \leq n} (\lfloor n/k \rfloor!)^k$, for $1 \leq n \leq 40$; (iii) confirms the conjecture directly, that $d(n)$ divides $G(n)$, for $1 \leq n \leq 40$; (iv) confirms Theorem 4 for every $1 \leq n \leq 50$ and every prime $p \leq n$, that is 437 pairs (n, p) , with 0 mismatches, and prints $B(0), \dots, B(12) = 0, 0, 0, 1, 1, 4, 3, 8, 8, 12, 13, 22, 17$; and (v) confirms that the quotient $G(n)/d(n)$ reproduces the published DATA of A129365, 1, 1, 1, 1, 1, 2, 2, 2, 6, 48, 48, 48.

References

- [1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A129364>, consulted 24 August 2026; also A066841, A092287, A129365 and A004125.
- [2] A. M. Legendre, *Théorie des Nombres*, 3rd ed., Firmin Didot, Paris, 1830.

A Suggested OEIS comment (plain text, house style)

The conjecture is true, with an exact exponent. Let $B = A004125$, so that $B(M) = \text{Sum}_{\{k=1..M\}} (M \bmod k)$. Then for every prime p ,

$$v_p(A092287(n)) - v_p(a(n)) = \text{Sum}_{\{i \geq 1\}} B(\text{floor}(n/p^i)).$$

Proof: $v_p(\gcd(j, k)) = \min(v_p(j), v_p(k))$ counts the $i \geq 1$ with p^i dividing both j and k , so summing over the square $[1, n]^2$ and exchanging the order of summation gives $v_p(A092287(n)) = \text{Sum}_i \text{floor}(n/p^i)^2$. For $a(n)$, use the formula $a(n) = \text{Product}_{\{k=1..n\}} (\text{floor}(n/k)!)^k$ together with Legendre and $\text{floor}(n/(k \cdot p^i)) = \text{floor}(\text{floor}(n/p^i)/k)$ to get $v_p(a(n)) = \text{Sum}_i \text{Sum}_{\{k=1..M_i\}} k \cdot \text{floor}(M_i/k)$, where $M_i = \text{floor}(n/p^i)$. Since $M^2 = \text{Sum}_{\{k=1..M\}} M$, the level- i difference is $\text{Sum}_{\{k=1..M_i\}} (M_i - k \cdot \text{floor}(M_i/k)) = \text{Sum}_{\{k=1..M_i\}} (M_i \bmod k) = B(M_i)$.

Every remainder is ≥ 0 , so every exponent is ≥ 0 and $a(n)$ divides $A092287(n)$. The quotient is A129365, and this is exactly Conjecture A of that entry - the two conjectures are the same statement, and one proof settles both.